

RENEAL TANZANIA

Lab Usage vs ICT NECTA Results

Analytics Dashboard — Description & Purpose

CSEE Computer Studies · Exam Code 036

Form 2 (Kidato cha Pili) & Form 4 (Kidato cha Nne) · 2023 – 2025

Internal analytics tool · Reneal Tanzania /box intranet

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1. Overview

The Lab Usage vs NECTA Results analytics dashboard is a web application built for the Reneal Tanzania internal team. It is hosted on the Reneal /box intranet and is accessible to the internal members to support the network of government secondary school computer labs across Tanzania.

The page brings together two data sources — student system usage hours recorded by the Reneal Reporter tool, and CSEE Computer Studies (Exam 036) grade results extracted from the national NECTA results — to answer one central question:

Does the amount of time students spend using the Reneal computer lab system have a measurable relationship with how well they perform in the national ICT examination?

The dashboard covers the academic years 2023, 2024, and 2025, and tracks both Form 2 (Kidato cha Pili) and Form 4 (Kidato cha Nne) cohorts across 36 matched schools — schools that appear in both the usage records and the NECTA results dataset.

2. Purpose

The dashboard serves four distinct purposes for Reneal Tanzania:

2.1 Measuring lab impact

Reneal's primary mission is to improve educational outcomes in underserved Tanzanian secondary schools by installing and maintaining solar-powered computer labs. The dashboard provides the first systematic attempt to measure whether this investment is translating into improved exam results. By plotting usage hours against NECTA pass rates for each school-year combination, the team can see — with real data rather than anecdote — whether higher lab engagement is associated with stronger academic performance.

2.2 Identifying schools that need support

Not all schools respond to lab access in the same way. The quadrant analysis groups schools into four categories based on their position relative to the median usage and median pass rate. Schools in the "High usage, low pass" and "Low usage, low pass" quadrants are highlighted as priority intervention targets — these are schools where the lab is either not being used effectively or where both access and outcomes are poor. The dashboard makes it straightforward for the country director to see which schools need a field visit, additional teacher training, or a curriculum-alignment review.

2.3 Informing donor and board reporting

Reneal regularly reports to donors and its board on the impact of its school installations. The correlation analysis, year-on-year trend charts, and top/bottom performer tables provide ready-made evidence for these reports. The Pearson r statistic and the quadrant breakdown give a statistically grounded, honest assessment of the relationship between lab access and exam results — one that the team can cite with confidence.

2.4 Tracking network growth and data quality over time

The "students who sat" bar chart and the year-on-year trend panel reveal how Reneal's data collection has matured alongside the school network. In 2023 only 13 records were matched; by 2025 this grew to 45 records covering 1,414 students who sat the exam. This growth story — from a handful of established schools to a broad multi-region network — is visible in the dashboard and helps the team understand both the expanding reach of the labs and where data gaps still exist.

3. Data Sources

The dashboard draws on two primary data files:

1. <https://matokeo.necta.go.tz>
2. Internal data source – Web To Sync Files